Type MS · Velocity and acceleration by differentiating

Answer 1 of 12 · Question 3 part a · 2 marks

2019 JUN · Paper 02 · extract 1 of 11 · find part a in this mark scheme extract

3(a)	$\frac{dv}{dt} = 2t - 4$ Accel $= 2\ \left(m/s^2\right)$	M1 A1 (2)
(b)	$s = \int_0^6 \left(t^2 - 4t + 7\right)dt = \left[\frac{t^3}{3} - 2t^2 + 7t\right]_0^6$ $= \frac{6^3}{3} - 2\times 6^2 + 7\times 6 = 42\ (m)$	M1A1 dM1 A1cao (4) [6]

(a)	
M1	Attempt to differentiate the expression for v. Power of t to decrease in at least one term and increase in none.
A1	Substitute $t = 3$ and obtain correct acceleration – units may be missing
(b)	
M1	Attempt to integrate the expression for v. Power of t to increase in at least one term and decrease in none. Ignore limits if shown. Constant not needed for indefinite integration.
A1	Correct integration. Limits/constant not needed.
dM1	**Either** substitute the limits 0 and 6 **or** use $s = 0$, $t = 0$ to obtain a value for the constant and substitute $t = 6$ in the complete expression. (Substitution of 0 can be implied if the result would have been 0) Depends on the previous M mark If more values of t are substituted **and results used** award M0
A1cao	$S = 42$ (m)
NB	Ans 42 w/o working scores 4/4 (Done on a calculator)

Type MS · Velocity and acceleration by differentiating

Answer 1 of 12 · Question 3 part a · 2 marks

2019 JUN · Paper 02 · extract 2 of 11 · find part a in this mark scheme extract

Question Number	Scheme	Marks
5	$(x+2y=17)\quad x=\frac{36}{y}\quad \left(\text{or } y=\frac{36}{x}\right)$	M1
	$\frac{36}{y}+2y=17,\quad 36+2y^2=17y\quad (\text{or } 72+x^2=17x)$	M1
	$2y^2-17y+36\ (=0)\qquad (\text{or } x^2-17x+72=0)$	A1
	$(y-4)(2y-9)=0\qquad (\text{or } (x-8)(x-9))$	
	$y=4\quad x=9$	dM1A1
	$y=4\frac{1}{2}\quad x=8$	A1 (6)
M1	Rearrange $xy = 36$ to $x = ...$ or $y = ...$	
M1	Eliminate x or y from the linear equation and obtain a 3TQ, $=0$ not needed	
A1	Correct 3TQ, terms in any order. $=0$ not needed	
M1	Solve their 3TQ by any valid method. Obtain at least one value for y or x	
A1	**Either** 2 correct values for x or y **or** a correct (x, y) pair	
A1	Both pairs correct and pairing clear.	
ALT:	The following method may possibly be seen:	
	$xy + x + 2y = 53 \Rightarrow 36 + x + 2y = 53 \Rightarrow x + 2y = 17$ and $xy = 36$ or $x \times 2y = 72$	M1
	Hence x and $2y$ are the roots of the equation $z^2 - 17z + 72 = 0$	M1A1
	$(z - 9)(z - 8) = 0 \Rightarrow z = 9$ or 8	M1
	So $x = 8$ $y = 4.5$ or $x = 9$ $y = 4$	A1A1 [6]
M1	Substitute $xy = 36$ in the linear equation to obtain $x + 2y = 17$ and $xy = 36$ oe	
M1	Obtain a 3TQ with roots x and $2y$	
A1	Correct 3TQ	
M1	Solve their 3TQ by any valid method. Obtain at least one value for *for the roots*	
A1	**Either** 2 correct values for x or y **or** a correct (x, y) pair	
A1	Both pairs correct and pairing clear.	
	Special Case $x+2y=17\quad xy=36$ Use $xy=36$ in the other equation to obtain $x+2y=17$ $\Rightarrow x=9\quad y=4$ By inspection: Score M1M0A0M1A1A0 (Must see $x+2y=17$; otherwise no marks) If the second answer is also obtained correctly by inspection, award all marks	

Type MS · Velocity and acceleration by differentiating

Answer 1 of 12 · Question 3 part a · 2 marks

2019 JUN · Paper 02 · extract 3 of 11 · find part a in this mark scheme extract

Question Number	Scheme	Marks
6(a)(i)	$\frac{dy}{dx} = 4e^{2x} + 2(4x-3)e^{2x}$	M1A1A1 (3)
(ii)	$(4x-3)\frac{dy}{dx} = (4x-3)(8x-2)e^{2x} = (8x-2)y$ *	M1A1cso (2)
(b)	$\frac{dy}{dx} = \frac{5\cos 5x \times (x-3)^2 - \sin 5x \times 2(x-3)}{(x-3)^4}$	M1A1A1 (3)
ALT	Using product rule:	
	$y = (x-3)^{-2}\sin 5x$	M1
	$\frac{dy}{dx} = -2(x-3)^{-3}\sin 5x + 5(x-3)^{-2}\cos 5x$	A1A1
		[8]

(a)(i)

M1 Use product rule to differentiate the given expression. Must have 2 terms added. One to be of the form ke^{2x} and the other of the form $k'(4x-3)e^{2x}$ where k' = 1 or 2

A1 Either term correct

A1 Second term correct

NB No simplification needed for these 3 marks

ALT $y = 4xe^{2x} - 3e^{2x} \Rightarrow \frac{dy}{dx} = 4e^{2x} + 8xe^{2x} - 6e^{2x}$

M1 Expand the given expression and differentiate using the product rule for $4xe^{2x}$

A1 Any 2 terms correct; A1 Third term correct.

NB No simplification needed for these 3 marks

(ii)

M1 Use their result from (i) to obtain an expression for $(4x-3)\frac{dy}{dx}$. No need to simplify.

A1cso Correct **given** result obtained with no errors in the working.
Can start with LHS and show equal to the RHS or vice versa or can start with each side and "meet in the middle"

(b)

M1 Attempt the quotient rule. The denominator must be $(x-3)^4$ and the numerator must be of the form $\left(k\cos 5x \times (x-3)^2 - \sin 5x \times l(x-3)\right)$ $k = \pm 5$ or ± 1, $l = 1$ or 2
(ie sine may have been differentiated to − cosine)

A1 One fully correct term in numerator.

A1 All fully correct.

ALT

M1 Rewrite without a quotient and apply the product rule obtaining 2 terms of the form shown

A1 Either term correct

A1 Second term correct
No need to simplify

Type MS · Velocity and acceleration by differentiating

Answer 1 of 12 · Question 3 part a · 2 marks

2019 JUN · Paper 02 · extract 4 of 11 · find part a in this mark scheme extract

Question Number	Scheme	Marks
7(a)	(i) $a = 9$	B1
	(ii) $d = 4$	B1 (2)
(b)	(i) $a = 4$	B1
	(ii) $r = 3$	B1 (2)
(c)	$A_{14} = \frac{14}{2}(2\times 9+13\times 4)$ or $\frac{14}{2}(9+61)$, $= 490$	M1, A1
	$"490"-6 = \frac{4(3^n-1)}{3-1}$	M1
	$3^n = 243$ $n = 5$	ddM1A1 (5) [9]

(a)
B1 Correct value, no working or explanation needed
B1 Correct value, no working or explanation needed
(b)
B1 Correct value, no working or explanation needed
B1 Correct value, no working or explanation needed
(c)
M1 Use either formula for the sum of an arithmetic series with their a and d (if needed) and obtain a value for the sum of the first 14 terms
A1 Correct value for the sum
M1 Subtract 6 from their sum (explicitly or implicitly) and equate to the sum of the first n terms of the geometric series obtained using their a and r
ddM1 Solve their equation by a correct method. No method need be shown but must reach n = ... Depends on both M marks above
A1 Correct value for n obtained

ALT For the last 3 marks:
M1 Subtract 6 from their sum and generate at least the first 5 terms of the geometric series.
ddM1 Sum their terms until at least "484" is reached
A1 Correct answer (5) obtained from correct work.

Type MS · Velocity and acceleration by differentiating

Answer 1 of 12 · Question 3 part a · 2 marks

2019 JUN · Paper 02 · extract 5 of 11 · find part a in this mark scheme extract

Question Number	Scheme	Marks
8(a)	$AB^2 = 4^2 + 2^2,\ BC^2 = 2^2 + 6^2,\ AC^2 = 2^2 + 4^2$	M1 (any one)
	(i) $AB = \sqrt{20}$ (4.47) (ii) $BC = \sqrt{40}$ (6.32) (iii) $AC = \sqrt{20}$ (4.47) or equivalents	A1A1A1 (4)
(b)	Any complete method for finding one of the angles: eg $AB^2 + AC^2 = BC^2 \Rightarrow \angle A = 90°$ or use trigonometry	M1
	$\angle A = 90°,\ \angle B = \angle C = 45°$	A1, A1 (3)
I	(centre at midpoint of BC) $(5, 5)$	M1A1 (2)
(d)	Radius $= \frac{1}{2}BC = \frac{1}{2}\sqrt{40} = \sqrt{10}$ (Working for (d) may be seen in a previous part)	M1A1 (2) [11]

(a)	
M1	Use Pythagoras with a plus sign to obtain AC^2, BC^2 or AC^2. If the answer is incorrect it must be clear that the correct coordinates have been used correctly.
A1A1A1	Award A1 for each correct length. Ignore labels (i), (ii) and (iii). Award M1A1A1A1 / M1A1A1A0 / M1A1A0A0 as appropriate.
SC:	If there is no working shown but at least one length is correct, award M1 and deduct one A mark for each incorrect length. (no length correct and no working ⇒ M0) If all 3 lengths are correct to at least 3 sf, award M1A1A1A0 If 2 are correct to at least 3 sf, award M1A1A0A0
(b)	
M1	Attempt to obtain any of the required angles. Method must be complete (ie reach a value for one angle) and formula used must be correct and values must be substituted into a correct formula.
A1	$\angle A = 90°$ Any labelling given can be ignored.
A1	$\angle B = \angle C = 45°$ All 3 correct w/o working scores M1A1A1
(c)	
M1	For indicating that the centre is at the midpoint of BC. This can be stated explicitly or used by attempting to find the midpoint. **OR:** Find equations for perpendicular bisectors of 2 of the sides and find the point of intersection
A1	Both coordinates correct. Correct answer written down w/o working scores M1A1
(d)	
M1	For indicating that the radius is half the length of BC. This can be stated explicitly or used by attempting to find half of their BC (not nec in the required form).
A1	Correct length of the radius, in the required form.
NB	If half the length of BC has been found earlier the marks for (d) can only be awarded if the length of the radius has been written in (d).

Type MS · Velocity and acceleration by differentiating

Answer 1 of 12 · Question 3 part a · 2 marks

2019 JUN · Paper 02 · extract 6 of 11 · find part a in this mark scheme extract

Question Number	Scheme	Marks
9(a)	$f(x)=\frac{1}{4}x^4-\frac{1}{3}x^3-2x^2+4x\ (+c)$	M1A1
	$x=-2\ \ y=-\frac{28}{3}\ \Rightarrow c=0$	M1
	$\left(f(x)=\frac{1}{4}x^4-\frac{1}{3}x^3-2x^2+4x\right)$ $\therefore C$ passes through O	A1 cso (4)
(b)(i)	$x=2\ \ f'(x)=8-4-8+4=0$	M1
	$\frac{d^2y}{dx^2}=3x^2-2x-4$	M1
	$x=2\quad \frac{d^2y}{dx^2}=12-4-4>0\quad \therefore$ min at $x=2$	A1cso
	$x=1\ \ f'(x)=1-1-4+4=0$	M1
	$x=1\quad \frac{d^2y}{dx^2}=3-2-4<0\quad \therefore$ max at $x=1$	A1 cso
ALT	$f'(x)=(x-2)(x-1)(x+2)\ (=0)$ factorise	M1
	$x=2,1,(-2)$ solve (solutions to be 2,1 (and another))	M1
	OR: $f'(x)\ (=0)$ solved by calculator.	
	All 3 solutions needed (and correct) $=0$ not needed M2	
	$\frac{d^2y}{dx^2}=3x^2-2x-4$ differentiate	M1
	$x=2\quad \frac{d^2y}{dx^2}=12-4-4>0\quad \therefore$ min at $x=2$	A1cso
	$x=1\quad \frac{d^2y}{dx^2}=3-2-4<0\quad \therefore$ max at $x=1$	A1cso
(ii)	$x=1\Rightarrow y=1\frac{11}{12}\qquad x=2\Rightarrow y=1\frac{1}{3}$	B1B1 (7)
(c)	$y'=(x-1)(x-2)(x+2)$	M1
(i)	$x=-2,\ y=-\frac{28}{3}$ or $\left(-2,-\frac{28}{3}\right)$	A1
(ii)	$x=-2\quad \frac{d^2y}{dx^2}=12+4-4>0\quad \therefore$ min point	A1cso (3)
		[14]

Type MS · Velocity and acceleration by differentiating

Answer 1 of 12 · Question 3 part a · 2 marks

2019 JUN · Paper 02 · extract 7 of 11 · find part a in this mark scheme extract

Question Number	Scheme	Marks
(a)M1	Attempt to integrate f'(x). The power of at least one x term must increase and none should decrease. c not needed	
A1	Correct integration, c not needed	
M1	Substitute the given coordinates to show $c = 0$. If c is not included (or assumed to be 0), then showing that substitution of $x = -2$ gives $y = -28/3$ is acceptable. Substitutions must be shown.	
A1cso	Correct conclusion from fully correct work. Accept eg $f(0) = 0$ ∴ shown	
(b)	Ignore labels (i) and (ii) when marking (b)	
(i)M1	Substitute $x = 2$ in the expression for f'(x) to show f'(x) = 0. Substitution must be shown	
M1	Differentiate the expression for f'(x). At least one power must decrease and none increase.	
A1cso	Show second derivative is > 0 at $x = 2$ and give the conclusion. No errors or omissions in the working.	
M1	Substitute $x = 1$ in the expression for f'(x) to show f'(x) = 0. Substitution must be shown	
A1cso	Show second derivative is < 0 at $x = 1$ and give the conclusion. No errors or omissions in the working.	
(ii)B1	For either y coordinate correct (and x coordinate correctly indicated; substitution shown indicates this)	
B1	For the second y coordinate correct	
(c)	(May have been seen in (b))	
M1	Factorise f'(x) completely – any valid method OR use the factor theorem to find $x = -2$	
(i)A1	Extract the x coordinate of the third turning point and obtain the corresponding y coordinate. May quote y coordinate from the question	
(ii)A1cso	Test the sign of the second derivative at this point and make the conclusion. All work in (c) and $\frac{d^2y}{dx^2}$ (from (b)) must be completely correct for this mark to be awarded.	
	Alternative ways to determine the nature of the turning points:	
1.	If the change of sign of f'(x) is used then values of f'(x) either side of 1 and 2 must be calculated to provide evidence.	
2.	The continuity of a cubic function can be used to establish the nature of the turning points. If in doubt send to review.	

Type MS · Velocity and acceleration by differentiating

Answer 1 of 12 · Question 3 part a · 2 marks

2019 JUN · Paper 02 · extract 8 of 11 · find part a in this mark scheme extract

Question Number	Scheme	Marks
10(a)	$\alpha+\beta=-3 \quad \alpha\beta=-5$	B1
(i)	$\alpha^2+\beta^2=(\alpha+\beta)^2-2\alpha\beta, \ =19$	M1,A1
(ii)	$\alpha^4+\beta^4=(\alpha^2+\beta^2)^2-2\alpha^2\beta^2, \ =19^2-50=311$	M1,A1(5)
	OR: $\alpha^4+\beta^4=(\alpha+\beta)^4-4\alpha\beta(\alpha^2+\beta^2)-6\alpha^2\beta^2, \ =19^2-50=311$	
(b)	$(\alpha-\beta)^2=\alpha^2-2\alpha\beta+\beta^2=19+10$ OR $(\alpha-\beta)^2=(\alpha+\beta)^2-4\alpha\beta=9-(-20)$	M1
	$\alpha-\beta=\sqrt{29}$ *	A1 cso (2)
(c)	$\alpha^4-\beta^4=(\alpha^2-\beta^2)(\alpha^2+\beta^2)=(\alpha-\beta)(\alpha+\beta)(\alpha^2+\beta^2)$	M1A1A1 (3)
(d)	$\alpha^4-\beta^4=\sqrt{29}\times(-3)\times19=-57\sqrt{29} \ \left(-\sqrt{94221}\right)$	M1A1 (2)
(e)	$2\beta^4=\alpha^4+\beta^4-(\alpha^4-\beta^4)$	M1
	$\beta^4=\frac{1}{2}(311+57\sqrt{29}),=\frac{311}{2}+\frac{57}{2}\sqrt{29}$	A1,A1 (3)
	$p=\frac{311}{2} \quad q=\frac{57}{2}$	[15]
ALT	$\beta^4=\left(\frac{-3-\sqrt{29}}{2}\right)^4$ **and** use a correct binomial expansion M1A1	
	Correct final answer A1	

(a)B1	Correct sum **and** product of roots, seen explicitly or used (in (a)). Must be clear that sum is negative
(i)M1	Correct algebra, ready for substitution of sum and product
A1	Correct answer, condone use of $\alpha+\beta=3$.
(ii)M1	Correct algebra, ready for substitution
A1	Correct answer, condone use of $\alpha+\beta=3$.
(b)M1	Correct algebra and substitution of their values
A1cso	Correct answer from correct working. Must have seen sum = -3 here if not shown in (a)
(c)	
M1	Factorise to 2 quadratic brackets or 2 linear and one quadratic bracket
A1	Obtain 2 linear and 1 quadratic brackets with 2 of the 3 brackets correct
A1	Third correct bracket Accept $(\alpha^2+\beta^2)$ or $((\alpha+\beta)^2-2\alpha\beta)$
(d)M1	Substitute their values for each of the 3 brackets obtained in (c)
A1	Correct answer as shown or equivalent **exact** value
(e)	
M1	Correct expression for $2\beta^4$ or β^4
A1ft	Substitute their numbers to obtain a numerical expression for β^4 The expression must be exact but need not be simplified **NB** A correct numerical expression for their values implies M1
A1	Correct answer in the required form. p and q need not be shown explicitly.

Type MS · Velocity and acceleration by differentiating

Answer 1 of 12 · Question 3 part a · 2 marks

2019 JUN · Paper 02 · extract 9 of 11 · find part a in this mark scheme extract

Question Number	Scheme	Marks
11(a)	$AC = \sqrt{(16^2 + 16^2)} = 16\sqrt{2}$	M1A1
ALT:	$AP^2 + PD^2 = 16^2 \Rightarrow AP = 8\sqrt{2}$	
	$VP = 8\sqrt{2}\tan 45 = 8\sqrt{2}$ (where P is the centre of the base) *	M1A1cso (4)
(b)	$VA^2 = \left(8\sqrt{2}\right)^2 + \left(8\sqrt{2}\right)^2 (= 256)$ or $VA = \dfrac{8\sqrt{2}}{\sin 45°}$	M1A1
	$VA = 16$ cm	A1 (3)
(c)	$DX^2 = 16^2 - 8^2$ where X is the foot of the perpendicular from D to VA	M1A1
	$DX = 8\sqrt{3}$	A1 (3)
(d)	$\tan\theta = \dfrac{8\sqrt{2}}{8}, \sin\theta = \dfrac{8\sqrt{2}}{8\sqrt{3}}, \cos\theta = \dfrac{8}{8\sqrt{3}}$	M1A1
	(or unsimplified if cosine or sine rule used) $\theta = 54.7°$	A1 (3)
(e)	$\cos\phi = \dfrac{\left(8\sqrt{3}\right)^2 + \left(8\sqrt{3}\right)^2 - \left(16\sqrt{2}\right)^2}{2\times 8\sqrt{3}\times 8\sqrt{3}} \left(= -\dfrac{1}{3}\right)$	M1A1
	$\phi = 109.5°$	A1 (3) [16]

(a)
M1 Use Pythagoras (with a + sign) to obtain the length of the diagonal of the base. Or use Pythagoras with correct sign to obtain the half diagonal
A1 Correct length for the diagonal or half diagonal
M1 Use tan in ΔAPV, their AP and angle of 45° to obtain the height
A1cso Correct answer with no errors in the working
OR: State ΔAVP is isosceles, or shown the 2 correct angles – can be on a diagram M1
$VP(= AP) = 8\sqrt{2}$ A1
OR use any other complete valid method M1 Correct result A1

(b)
M1 Use Pythagoras or trigonometry in ΔAPV or ΔAVC (or any other complete, valid method) to obtain a numerical expression for VA.
A1 Correct numbers in their choice of method.
A1 Correct length obtained.
$AV = 16$ w/o working scores M1A1A1

Type MS · Velocity and acceleration by differentiating

Answer 1 of 12 · Question 3 part a · 2 marks

2019 JUN · Paper 02 · extract 10 of 11 · find part a in this mark scheme extract

Question Number	Scheme	Marks
(c)		
M1	Use Pythagoras with a minus sign (seen or implied) or trigonometry in ΔADX OR any other complete valid method **NB**: triangle ΔADV is equilateral.	
A1	Correct numbers in their choice of method.	
A1	Correct **exact** length for the perpendicular.	
(d)		
M1	Identify the correct triangle needed with the required angle marked (may be on Figure 1). This may be shown explicitly or implied by their work that follows.	
A1	Reach one of $\tan\theta = \frac{8\sqrt{2}}{8}$, $\sin\theta = \frac{8\sqrt{2}}{8\sqrt{3}}$, $\cos\theta = \frac{8}{8\sqrt{3}}$ oe	
A1	Correct answer, **must be 1 dp.**	
(e)		
M1	Use cosine rule: $\cos\phi = \frac{"DX"^2 + "XB"^2 - "BD"^2}{2\times"DX"\times"XB"}$ (their values)	
A1	Correct numbers substituted, follow through their previous answers	
A1	Correct answer, **must be 1 dp unless already penalised in (d)**	
	Any other routes should be marked: M1 Correct, complete method (ie it must be possible to reach a value for the required angle) A1 Correct numbers substituted A1 Correct answer, **must be 1 dp unless already penalised in (d)**	

Type MS · Velocity and acceleration by differentiating

Answer 1 of 12 · Question 3 part a · 2 marks

2019 JUN · Paper 02 · extract 11 of 11 · find part a in this mark scheme extract

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Type MS · Velocity and acceleration by differentiating

Answer 2 of 12 · Question 1 part b · 2 marks

2025 JUN · Paper 01 · extract 1 of 2 · find part b in this mark scheme extract

1 (a)	$b^2-4ac=(-9)^2-4\times3\times7=-3$ $-3<0$ so no (real) solutions for v	M1 A1 [2]
(b)	$\left\{\frac{dv}{dt}\right\}=6t-9\Rightarrow a=6\times4-9=15\left(m/s^2\right)$	M1A1 [2]
(c)	$s=\int_0^5(3t^2-9t+7)\,dt=\left[t^3-\frac{9}{2}t^2+7t\right]_0^5$ $=5^3-\frac{9}{2}\times5^2+7\times5=\frac{95}{2}$(m) oe	M1 M1A1 [3]
		Total 7 marks

Part	Mark	Additional Guidance
(a)	M1	Applies the discriminant, or any other valid method. e.g. Attempts to solve the 3TQ by quadratic formula, Completing square, achieves $3\left(t-\frac{3}{2}\right)^2+...$ Attempts to solve $v = 0$, finds complex roots
	A1	Verifies that the equation for v has no (real) solutions By completing square, it must be correct, $3\left(t-\frac{3}{2}\right)^2+\frac{1}{4}$, states the graph is above x axis, or > 0 Finds correct complex roots $\frac{9\pm\sqrt{3}i}{6}$, states imaginary numbers/roots **Note**: Just stating "error" is not enough
(b)	M1	Attempts to differentiate the expression for v. Power of t to decrease in at least one term and increase in none.
	A1	Substitutes $t = 4$ and obtains correct acceleration, with or without unit.
(c)	M1	Attempt to integrate the expression for v. Power of t to increase in at least one term and decrease in none. Ignore limits if shown.
	M1	Clear substitution of the **given** limits into their **integrated** expression for distance, no need to see substitution of zero, can be implied by the correct distance found from a **correct integrated** expression
	A1	$\frac{95}{2}$ (m) oe e.g. 47.5, ignore unit

Type MS · Velocity and acceleration by differentiating

Answer 2 of 12 · Question 1 part b · 2 marks

2025 JUN · Paper 01 · extract 2 of 2 · find part b in this mark scheme extract

Question number	Scheme	Marks

Type MS · Velocity and acceleration by differentiating

Answer 3 of 12 · Question 6 part a · 2 marks

2024 JUN · Paper 01 (regional R) · find part a in this mark scheme extract

Question	Scheme	Marks
6(a)	$t=0 \Rightarrow s=e^{2\times 0}\sin\left(3\times 0\right)+2=2$ $t=\frac{\pi}{6} \Rightarrow s=e^{2\times\frac{\pi}{6}}\sin\left(3\times\frac{\pi}{6}\right)+2=e^{\frac{\pi}{3}}+2$ $AB=e^{\frac{\pi}{3}}+2-2=e^{\frac{\pi}{3}}$ (m)	M1 A1 [2]
(b)	$v=2e^{2t}\sin 3t+3e^{2t}\cos 3t$ When $t=\frac{\pi}{3}$ $v=2e^{\frac{2\pi}{3}}\sin\left(\frac{\pi}{3}\times 3\right)+3e^{\frac{2\pi}{3}}\cos\left(\frac{\pi}{3}\times 3\right)=-3e^{\frac{2\pi}{3}}$ (m/s)	M1A1A1 B1ft [4]
		Total 6 marks

Type MS · Velocity and acceleration by differentiating

Answer 4 of 12 · Question 6 part b · 4 marks

2024 JUN · Paper 01 (regional R) · find part b in this mark scheme extract

Question	Scheme	Marks
6(a)	$t=0 \Rightarrow s=e^{2\times 0}\sin(3\times 0)+2=2$ $t=\frac{\pi}{6} \Rightarrow s=e^{2\times\frac{\pi}{6}}\sin\left(3\times\frac{\pi}{6}\right)+2=e^{\frac{\pi}{3}}+2$ $AB=e^{\frac{\pi}{3}}+2-2=e^{\frac{\pi}{3}}$ (m)	M1 A1 [2]
(b)	$v=2e^{2t}\sin 3t+3e^{2t}\cos 3t$ When $t=\frac{\pi}{3}$ $v=2e^{\frac{2\pi}{3}}\sin\left(\frac{\pi}{3}\times 3\right)+3e^{\frac{2\pi}{3}}\cos\left(\frac{\pi}{3}\times 3\right)=-3e^{\frac{2\pi}{3}}$ (m/s)	M1A1A1 B1ft [4]
		Total 6 marks

Type MS · Velocity and acceleration by differentiating

Answer 5 of 12 · Question 2 part a · 2 marks

2022 JUN · Paper 01 (regional R) · extract 1 of 5 · find part a in this mark scheme extract

2(a)	$(a=)2t-3$	M1 A1 [2]
(b)	$(s=\int(t^2-3t+4)\,dt)$	
	$(s=)\frac{t^3}{3}-\frac{3}{2}t^2+4t(+C)$ oe	M1 A1
	$(t=2, s=7)$	
	$\frac{2^3}{3}-\frac{3}{2}\times2^2+4\times2+C=7\Rightarrow C=\frac{7}{3}$	dM1
	$\frac{4^3}{3}-\frac{3}{2}\times4^2+4\times4+'\frac{7}{3}'=\frac{47}{3}$ oe	M1A1
		[5]
	ALT	
	Displacement $-7=\int_2^4\left(x^2-3x+4\right)dx$	
	$=\left[\frac{x^3}{3}-\frac{3x^2}{2}+4t\right]_2^4$	[M1A1
	$=\left[\frac{4^3}{3}-\frac{3\times4^2}{2}+4\times4\right]-\left[\frac{2^3}{3}-\frac{3\times2^2}{2}+4\times2\right]$	dM1M1
	$=\frac{47}{3}$	A1]
		Total 7 marks

Type MS · Velocity and acceleration by differentiating

Answer 5 of 12 · Question 2 part a · 2 marks

2022 JUN · Paper 01 (regional R) · extract 2 of 5 · find part a in this mark scheme extract

Part	Mark	Additional Guidance
2(a)	M1	For a minimally acceptable attempt at differentiation (see general guidance), no power of t must increase.
	A1	Correct expression with or without $a = \ldots$
(b)	M1	For a minimally acceptable attempt at integration (see general guidance), no power of t must decrease. + C is not necessary for this mark.
	A1	Fully correct integration. C is not necessary for this mark.
	dM1	Correct substitution of $t = 2$ into **their** integrated expression, correctly equated to 7 and an attempt to rearrange to find C. **Note: This mark is dependent on the first M mark in (b)**
	M1	Substitution of $t = 4$ correctly into their expression for s, provided it is a changed expression from v
	A1	Correct value oe must be an exact value.
	ALT	
	M1	For a minimally acceptable attempt at integration (see general guidance), no power of t must decrease.
	A1	Fully correct integration
	dM1	For the correct limits between $t = 4$ and $t = 2$ **and** equates to $d - 7$ or equivalent
	M1	Substitutes the limits to evaluate the integral
	A1	Correct value oe must be an exact value.

Type MS · Velocity and acceleration by differentiating

Answer 5 of 12 · Question 2 part a · 2 marks

2022 JUN · Paper 01 (regional R) · extract 3 of 5 · find part a in this mark scheme extract

Question number	Scheme	Marks

Type MS · Velocity and acceleration by differentiating

Answer 5 of 12 · Question 2 part a · 2 marks

2022 JUN · Paper 01 (regional R) · extract 4 of 5 · find part a in this mark scheme extract

Part	Mark	Additional Guidance
2(a)	M1	For a minimally acceptable attempt at differentiation (see general guidance), no power of t must increase.
	A1	Correct expression with or without $a = \ldots$
(b)	M1	For a minimally acceptable attempt at integration (see general guidance), no power of t must decrease. + C is not necessary for this mark.
	A1	Fully correct integration. C is not necessary for this mark.
	dM1	Correct substitution of $t = 2$ into **their** integrated expression, correctly equated to 7 and an attempt to rearrange to find C. **Note: This mark is dependent on the first M mark in (b)**
	M1	Substitution of $t = 4$ correctly into their expression for s, provided it is a changed expression from v
	A1	Correct value oe must be an exact value.
	ALT	
	M1	For a minimally acceptable attempt at integration (see general guidance), no power of t must decrease.
	A1	Fully correct integration
	dM1	For the correct limits between $t = 4$ and $t = 2$ **and** equates to $d - 7$ or equivalent
	M1	Substitutes the limits to evaluate the integral
	A1	Correct value oe must be an exact value.

Type MS · Velocity and acceleration by differentiating

Answer 5 of 12 · Question 2 part a · 2 marks

2022 JUN · Paper 01 (regional R) · extract 5 of 5 · find part a in this mark scheme extract

Question number	Scheme	Marks

Type MS · Velocity and acceleration by differentiating

Answer 6 of 12 · Question 4 part a · 2 marks

2023 JUN · Paper 01 · extract 1 of 3 · find part a in this mark scheme extract

Question	Scheme	Marks
4(a)	$a=\frac{dv}{dt}=4t-16 \quad [t=5] \quad a=4\times5-16=4$	M1A1 [2]
(b)	$v=0\Rightarrow 2t^2-16t+30=0\Rightarrow (t-5)(2t-6)=0$ $\Rightarrow t_1=3,\ t_2=5$	M1 A1A1
	Displacement $=\int_3^5 (2t^2-16t+30)\,dt=\left[\frac{2t^3}{3}-\frac{16t^2}{2}+30t\right]_3^5$	M1A1
	$=\left[\frac{2\times5^3}{3}-8\times25+30\times5\right]-\left[\frac{2\times3^3}{3}-8\times9+30\times3\right]=\frac{100}{3}-36=-\frac{8}{3}$	M1A1
	Distance $=\frac{8}{3}$ (m)	A1ft
		[8]
		Total 10 marks

Type MS · Velocity and acceleration by differentiating

Answer 6 of 12 · Question 4 part a · 2 marks

2023 JUN · Paper 01 · extract 2 of 3 · find part a in this mark scheme extract

Part	Mark	Notes
(a)	M1	For an attempt to differentiate the given expression and substitute $t = 5$ See General Guidance for the definition of an attempt. Do not accept an expression with any terms integrated.
	A1	Obtains the value 4
(b)	M1	Sets $v = 0$ and attempts to solve the equation using a correct method. Please check their work carefully. See General Guidance for acceptable ways to solve a quadratic. If they use a calculator and only $t = 3, 5$ are seen with no working – award M1A1A1
	A1	For either $t_1 = 3$ or $t_2 = 5$
	A1	For both $t_1 = 3$ and $t_2 = 5$
	No working seen to find the distance.	
	SC: If a final answer of $\frac{8}{3}$ is seen with no evidence of algebraic integration, **award the final A mark only.**	
	M1	For an attempt to integrate the given expression for v with or without limits. Ignore poor or absent notation. [ie. No integral sign] See General Guidance for the definition of an attempt. Note: Do not accept an integrated expression with any **algebraic** terms differentiated.
	A1	For the correct integrated expression unsimplified or simplified. isw incorrect simplification following a correctly integrated expression. Accept for this mark the inclusion of $+ c$ Ignore limits for this mark.
	M1	Substitutes their values of $t_1 = 3$ or $t_2 = 5$ into their integrated expression and attempts to evaluate. Allow the subtraction either way around. Their integral must be a changed expression. Accept even if they have differentiated it. If the final answer is not $\pm\frac{8}{3}$ and there is no explicit substitution seen, award M0.
	A1	For the correct value of $-\frac{8}{3}$ You must see the negative value for this mark. Accept awrt -2.67 for this mark.
	A1ft	For the correct distance of $\frac{8}{3}$ (m) or exact equivalent. Do not accept 2.67 (m) but accept $2.\dot{6}$ **Note:** This is a ft mark. We will only award this mark if their value for the displacement is **negative** and the final distance is positive.

Type MS · Velocity and acceleration by differentiating

Answer 6 of 12 · Question 4 part a · 2 marks

2023 JUN · Paper 01 · extract 3 of 3 · find part a in this mark scheme extract

Question	Scheme	Marks

Type MS · Velocity and acceleration by differentiating

Answer 7 of 12 · Question 5 part a · 2 marks

2023 JUN · Paper 02 (regional R) · extract 1 of 5 · find part a in this mark scheme extract

5(a)	$\frac{dv}{dt} = 4t - 19$ $\frac{dv}{dt} = 4\times 5 - 19 = 1\left(\text{m/s}^2\right)$	M1 A1 [2]
(b)	$2t^2 - 19t + 35 = 0 \Rightarrow (2t-5)(t-7) = 0 \Rightarrow t = \frac{5}{2}, 7$ $t_1 = \frac{5}{2} \quad t_2 = 7$	M1 A1 [2]
(c)	$D = \int_{\frac{5}{2}}^{7} \left(2t^2 - 19t + 35\right) dt = \left[\frac{2t^3}{3} - \frac{19t^2}{2} + 35t\right]_{\frac{5}{2}}^{7}$ $\left(\frac{2\times 7^3}{3} - \frac{19\times 7^2}{2} + 35\times 7\right) - \left(\frac{2\times 2.5^3}{3} - \frac{19\times 2.5^2}{2} + 35\times 2.5\right)$ $= \left[\frac{49}{6} - \frac{925}{24}\right] = [8.167 - 38.542] = \left[-\frac{243}{8} \text{ or } -30.375\right]$ $\Rightarrow D = \frac{243}{8}$ (m) oe e.g. 30.375	M1 M1 A1 [3]
		Total 7 marks

Type MS · Velocity and acceleration by differentiating

Answer 7 of 12 · Question 5 part a · 2 marks

2023 JUN · Paper 02 (regional R) · extract 2 of 5 · find part a in this mark scheme extract

Part	Mark	Notes
(a)	M1	For differentiating the given v and substituting in $t = 5$ into the derivative. Both differentiation and substitution must be correct for this mark.
	A1	For the correct acceleration of 1 m/s^2. Units are not required.
(b)	M1	For attempting to solve the given 3TQ for the velocity. A correct method must be used, and they must reach **two** values of t for the award of this mark. See General Guidance for the definition of an attempt to solve a 3TQ. If there is no visible method seen, both values of t must be seen for the award of this mark.
	A1	For the two correct values of t They do not need to ne identified as t_1 or t_2 for this mark. Accept $\frac{5}{2}$ and 7 seen.
(c)	M1	For an attempt to integrate the given expression for v. At least one term must be correct and no terms are to be differentiated. If the value of 35 'disappears' it is M0. The question states 'use calculus' so integration must be seen.
	M1	For substituting the **two** values of t into their integrated expression. and subtracting the result (either way) of both substitutions. Allow for any changed expression from v A correct answer of $\pm\frac{243}{8}$ or ± 30.375 implies correct substitution into a correct integral. If the integration is incorrect, or the values are incorrect, full correct substitution must be seen for the award of this mark.
	A1	For the correct distance (which must be a positive value).

Type MS · Velocity and acceleration by differentiating

Answer 7 of 12 · Question 5 part a · 2 marks

2023 JUN · Paper 02 (regional R) · extract 3 of 5 · find part a in this mark scheme extract

Question	Scheme	Marks

Type MS · Velocity and acceleration by differentiating

Answer 7 of 12 · Question 5 part a · 2 marks

2023 JUN · Paper 02 (regional R) · extract 4 of 5 · find part a in this mark scheme extract

Part	Mark	Notes
(a)	M1	For differentiating the given v and substituting in $t = 5$ into the derivative. Both differentiation and substitution must be correct for this mark.
	A1	For the correct acceleration of 1 m/s^2. Units are not required.
(b)	M1	For attempting to solve the given 3TQ for the velocity. A correct method must be used, and they must reach **two** values of t for the award of this mark. See General Guidance for the definition of an attempt to solve a 3TQ. If there is no visible method seen, both values of t must be seen for the award of this mark.
	A1	For the two correct values of t They do not need to ne identified as t_1 or t_2 for this mark. Accept $\frac{5}{2}$ and 7 seen.
(c)	M1	For an attempt to integrate the given expression for v. At least one term must be correct and no terms are to be differentiated. If the value of 35 'disappears' it is M0. The question states 'use calculus' so integration must be seen.
	M1	For substituting the **two** values of t into their integrated expression. and subtracting the result (either way) of both substitutions. Allow for any changed expression from v A correct answer of $\pm\frac{243}{8}$ or ± 30.375 implies correct substitution into a correct integral. If the integration is incorrect, or the values are incorrect, full correct substitution must be seen for the award of this mark.
	A1	For the correct distance (which must be a positive value).

Type MS · Velocity and acceleration by differentiating

Answer 7 of 12 · Question 5 part a · 2 marks

2023 JUN · Paper 02 (regional R) · extract 5 of 5 · find part a in this mark scheme extract

Question	Scheme	Marks

Type MS · Velocity and acceleration by differentiating

Answer 8 of 12 · Question 6 part a · 1 mark

2021 NOV · Paper 02 · extract 1 of 2 · find part a in this mark scheme extract

6 (a)	8 (m) (must be positive)	B1 [1]
(b)	$\frac{ds}{dt} = 3t^2 - 8t - 16$ $\frac{ds}{dt} = 0$ when P is stationary $\Rightarrow 3t^2 - 8t - 16 = 0$ $3t^2 - 8t - 16 = (3t + 4)(t - 4) = 0 \Rightarrow t = 4$	M1A1 M1A1 [4]
(c)	$\frac{d^2s}{dt^2} = 6t - 8$ $6t - 8 = 10 \Rightarrow t = 3$	B1ft M1A1 [3]
		Total 8 marks

Type MS · Velocity and acceleration by differentiating

Answer 8 of 12 · Question 6 part a · 1 mark

2021 NOV · Paper 02 · extract 2 of 2 · find part a in this mark scheme extract

Question number	Scheme	Marks

Type MS · Velocity and acceleration by differentiating

Answer 9 of 12 · Question 2 part a · 2 marks

2022 JUN · Paper 02 · extract 1 of 2 · find part a in this mark scheme extract

2 a	$v = \frac{dx}{dt} = 4t^3 - 13.5$	M1
	When $t = 3$ $v = 4(3)^3 - 13.5 = 94.5$ ms^{-1}	A1 (2)
b	$4t^3 - 13.5 = 0 \Rightarrow t^3 = \frac{27}{8} \Rightarrow t = 1.5$	M1 A1ft (2)
c	$a = \frac{dv}{dt} = 12t^2$	M1
	When $t = 2$ $a = 12 \times 2^2 = 48$ ms^{-2}	A1 (2)
		Total 6 marks

Part	Mark	Notes
Ignore incorrect/spurious notation through this question. e.g. ignore $\frac{dy}{dx} = ...$ or the LHS		
(a)	**M1**	For an attempt to differentiate the given expression [with no terms integrated] See General Guidance for the definition of an attempt to differentiate.
	A1	For substituting the value of $t = 3$ into their differentiated expression and obtains 94.5 (units not required)
(b)	**M1**	For setting their $\frac{dx}{dt} = 0$ which must be at a minimum of the form $\pm kt^3 \pm l$ and attempting to find a value for t
	A1ft	For $t = 1.5$ Ft their expression for v which must have come from an acceptable attempt to differentiate the given x
(c)	**M1**	For differentiating their $\frac{dx}{dt}$ which must be of the form $\frac{dx}{dt}$ (or v) $= \pm kt^3 \pm l$ where k and l are constants, [with no term integrated]
	A1	For substituting $t = 2$ into their differentiated expression and obtains 48 (units not required)

Type MS · Velocity and acceleration by differentiating

Answer 9 of 12 · Question 2 part a · 2 marks

2022 JUN · Paper 02 · extract 2 of 2 · find part a in this mark scheme extract

Question number	Scheme	Marks
	$\lvert\overrightarrow{OA}\rvert = \sqrt{\quad}$ $\lvert\overrightarrow{OB}\rvert = \sqrt{\quad}$	

Type MS · Velocity and acceleration by differentiating

Answer 10 of 12 · Question 2 part c · 2 marks

2022 JUN · Paper 02 · extract 1 of 2 · find part c in this mark scheme extract

2 a	$v = \frac{dx}{dt} = 4t^3 - 13.5$	M1
	When $t = 3$ $v = 4(3)^3 - 13.5 = 94.5$ ms^{-1}	A1 (2)
b	$4t^3 - 13.5 = 0 \Rightarrow t^3 = \frac{27}{8} \Rightarrow t = 1.5$	M1 A1ft (2)
c	$a = \frac{dv}{dt} = 12t^2$	M1
	When $t = 2$ $a = 12 \times 2^2 = 48$ ms^{-2}	A1 (2)
		Total 6 marks

Part	Mark	Notes
Ignore incorrect/spurious notation through this question. e.g. ignore $\frac{dy}{dx} = ...$ or the LHS		
(a)	**M1**	For an attempt to differentiate the given expression [with no terms integrated] See General Guidance for the definition of an attempt to differentiate.
	A1	For substituting the value of $t = 3$ into their differentiated expression and obtains 94.5 (units not required)
(b)	**M1**	For setting their $\frac{dx}{dt} = 0$ which must be at a minimum of the form $\pm kt^3 \pm l$ and attempting to find a value for t
	A1ft	For $t = 1.5$ Ft their expression for v which must have come from an acceptable attempt to differentiate the given x
(c)	**M1**	For differentiating their $\frac{dx}{dt}$ which must be of the form $\frac{dx}{dt}$ (or v) $= \pm kt^3 \pm l$ where k and l are constants, [with no term integrated]
	A1	For substituting $t = 2$ into their differentiated expression and obtains 48 (units not required)

Type MS · Velocity and acceleration by differentiating

Answer 10 of 12 · Question 2 part c · 2 marks

2022 JUN · Paper 02 · extract 2 of 2 · find part c in this mark scheme extract

Question number	Scheme	Marks
	$\left\lvert\overrightarrow{OA}\right\rvert = \sqrt{\quad}$ $\left\lvert\overrightarrow{OB}\right\rvert = \sqrt{\quad}$	

Type MS · Velocity and acceleration by differentiating

Answer 11 of 12 · Question 7 part a · 1 mark

2023 JAN · Paper 01 (regional R) · extract 1 of 3 · find part a in this mark scheme extract

7(a)	$v = 1^2 - 10\times1 + 28 = 19$ [m/s]	B1 [1]
(b)	$s = \int\left(t^2 - 10t + 28\right)\,\mathrm{d}t = \frac{t^3}{3} - \frac{10t^2}{2} + 28t\,(+c)$ $24 = \frac{3^3}{3} - \frac{10\times3^2}{2} + 28\times3 + c \Rightarrow c = -24 \Rightarrow \left[s = \frac{t^3}{3} - \frac{10t^2}{2} + 28t - 24\right]$ $t = 5, \quad s = \frac{5^3}{3} - \frac{10\times5^2}{2} + 28\times5 - 24 = \frac{98}{3}$ [m]	M1 M1A1 M1A1 [5]
(c)	$\frac{\mathrm{d}v}{\mathrm{d}t} = 2t - 10$ when $t = 9$, acceleration = 8 [m/s ^{2}]	M1 A1 [2]
(d)	(i) $v = (t-5)^2 + 3$ Irrespective of the value of t $v \geqslant 3$ so the particle never comes to rest. **ALT** $b^2 - 4ac < 0 \Rightarrow (-10)^2 - 4\times1\times28 = -12$ No real solutions so the particle never comes to rest. (ii) Least value of v is 3 [m/s]	M1 A1 [M1 A1] B1 [3]
		Total 11 marks

Question	Notes	Marks
7(a)	$v = t^2 - 10t + 28$	
	$v = 1^2 - 10\times1 + 28 = 19$ [m/s]	B1 [1]
(b)	For an attempt to integrate the given expression for v [See general guidance for the definition of an attempt] $s = \int\left(t^2 - 10t + 28\right)\,\mathrm{d}t = \frac{t^3}{3} - \frac{10t^2}{2} + 28t\,(+c)$	M1
	For finding the value of c. They cannot score this mark without $+c$ $24 = \frac{3^3}{3} - \frac{10\times3^2}{2} + 28\times3 + c \Rightarrow c = \ldots$	M1
	For the correct expression for s. This need not be explicitly stated. $s = \frac{t^3}{3} - \frac{10t^2}{2} + 28t - 24$	A1

Type MS · Velocity and acceleration by differentiating

Answer 11 of 12 · Question 7 part a · 1 mark

2023 JAN · Paper 01 (regional R) · extract 2 of 3 · find part a in this mark scheme extract

	Award for the correct value of c seen -24 [m]	
	For using their integrated expression for s to find a value of the displacement when $t = 5$ $s = \frac{5^3}{3} - \frac{10\times5^2}{2} + 28\times5 - 24 = ...$	M1
	For the correct value of $s = \frac{98}{3}$ [m] Accept $s = 32.7$ [or better]	A1 [5]
(c)	For an attempt to differentiate the given v and substituting $t = 9$ into their differentiated expression. $\frac{dv}{dt} = 2t - 10 \Rightarrow \frac{dv}{dt} = 2\times9 - 10 = ...$	M1
	For acceleration = 8 [m/s ^{2}]	A1 [2]
(d)(i)	**Method A** Completes the square to give $v = (t-5)^2 + 3$	M1
	Concludes that at the minimum velocity [3 m/s] $t = 5$ so P never comes to rest	A1
	Method B Finds the value of the discriminant $b^2 - 4ac < 0 \Rightarrow (-10)^2 - 4\times1\times28 = -12$	M1
	Concludes that as there are no real solutions, **so *P* does not come to rest.**	A1
	Method C Solves the 3TQ to give the following two [non-real] values of t: $5+\sqrt{3}\mathrm{i}$ and $5-\sqrt{3}\mathrm{i}$ or $\left(t-\left[5+\sqrt{3}\mathrm{i}\right]\right)\left(t-\left[5+\sqrt{3}\mathrm{i}\right]\right)=0$	M1
	Concludes that as there are no real solutions, **so *P* does not come to rest.**	A1
	Method D Uses their result from (c) $\frac{dv}{dt} = 2t - 10 = 0 \Rightarrow t = 5$	M1
	Concludes that at the minimum velocity [3 m/s] $t = 5$ so P never comes to rest	A1
(ii)	For the correct value of $v = 3$ [m/s]	B1 [3]
		Total 11 marks

Type MS · Velocity and acceleration by differentiating

Answer 11 of 12 · Question 7 part a · 1 mark

2023 JAN · Paper 01 (regional R) · extract 3 of 3 · find part a in this mark scheme extract

Question	Scheme	Marks

Type MS · Velocity and acceleration by differentiating

Answer 12 of 12 · Question 7 part c · 2 marks

2023 JAN · Paper 01 (regional R) · extract 1 of 3 · find part c in this mark scheme extract

7(a)	$v = 1^2 - 10\times1 + 28 = 19$ [m/s]	B1 [1]
(b)	$s = \int\left(t^2 - 10t + 28\right)\,\mathrm{d}t = \frac{t^3}{3} - \frac{10t^2}{2} + 28t\,(+c)$ $24 = \frac{3^3}{3} - \frac{10\times3^2}{2} + 28\times3 + c \Rightarrow c = -24 \Rightarrow \left[s = \frac{t^3}{3} - \frac{10t^2}{2} + 28t - 24\right]$ $t = 5, \quad s = \frac{5^3}{3} - \frac{10\times5^2}{2} + 28\times5 - 24 = \frac{98}{3}$ [m]	M1 M1A1 M1A1 [5]
(c)	$\frac{\mathrm{d}v}{\mathrm{d}t} = 2t - 10$ when $t = 9$, acceleration = 8 [m/s ^{2}]	M1 A1 [2]
(d)	(i) $v = (t-5)^2 + 3$ Irrespective of the value of t $v \geqslant 3$ so the particle never comes to rest. **ALT** $b^2 - 4ac < 0 \Rightarrow (-10)^2 - 4\times1\times28 = -12$ No real solutions so the particle never comes to rest. (ii) Least value of v is 3 [m/s]	M1 A1 [M1 A1] B1 [3]
		Total 11 marks

Question	Notes	Marks
7(a)	$v = t^2 - 10t + 28$	
	$v = 1^2 - 10\times1 + 28 = 19$ [m/s]	B1 [1]
(b)	For an attempt to integrate the given expression for v [See general guidance for the definition of an attempt] $s = \int\left(t^2 - 10t + 28\right)\,\mathrm{d}t = \frac{t^3}{3} - \frac{10t^2}{2} + 28t\,(+c)$	M1
	For finding the value of c. They cannot score this mark without $+c$ $24 = \frac{3^3}{3} - \frac{10\times3^2}{2} + 28\times3 + c \Rightarrow c = \ldots$	M1
	For the correct expression for s. This need not be explicitly stated. $s = \frac{t^3}{3} - \frac{10t^2}{2} + 28t - 24$	A1

Type MS · Velocity and acceleration by differentiating

Answer 12 of 12 · Question 7 part c · 2 marks

2023 JAN · Paper 01 (regional R) · extract 2 of 3 · find part c in this mark scheme extract

	Award for the correct value of c seen -24 [m]	
	For using their integrated expression for s to find a value of the displacement when $t = 5$ $s = \frac{5^3}{3} - \frac{10\times5^2}{2} + 28\times5 - 24 = ...$	M1
	For the correct value of $s = \frac{98}{3}$ [m] Accept $s = 32.7$ [or better]	A1 [5]
(c)	For an attempt to differentiate the given v and substituting $t = 9$ into their differentiated expression. $\frac{dv}{dt} = 2t - 10 \Rightarrow \frac{dv}{dt} = 2\times9 - 10 = ...$	M1
	For acceleration = 8 [m/s ^{2}]	A1 [2]
(d)(i)	**Method A** Completes the square to give $v = (t-5)^2 + 3$	M1
	Concludes that at the minimum velocity [3 m/s] $t = 5$ so P never comes to rest	A1
	Method B Finds the value of the discriminant $b^2 - 4ac < 0 \Rightarrow (-10)^2 - 4\times1\times28 = -12$	M1
	Concludes that as there are no real solutions, **so *P* does not come to rest.**	A1
	Method C Solves the 3TQ to give the following two [non-real] values of t: $5+\sqrt{3}\mathrm{i}$ and $5-\sqrt{3}\mathrm{i}$ or $\left(t-\left[5+\sqrt{3}\mathrm{i}\right]\right)\left(t-\left[5+\sqrt{3}\mathrm{i}\right]\right)=0$	M1
	Concludes that as there are no real solutions, **so *P* does not come to rest.**	A1
	Method D Uses their result from (c) $\frac{dv}{dt} = 2t - 10 = 0 \Rightarrow t = 5$	M1
	Concludes that at the minimum velocity [3 m/s] $t = 5$ so P never comes to rest	A1
(ii)	For the correct value of $v = 3$ [m/s}	B1 [3]
		Total 11 marks

Type MS · Velocity and acceleration by differentiating

Answer 12 of 12 · Question 7 part c · 2 marks

2023 JAN · Paper 01 (regional R) · extract 3 of 3 · find part c in this mark scheme extract

Question	Scheme	Marks